Bogleheads

Risk and return

The relationship between **risk** and **return** is a fundamental concept in finance theory, and is one of the most important concepts for investors to understand. A widely used definition of investment risk, both in theory and practice, is the uncertainty that an investment will earn its expected rate of return.[1] Finance literature uses "return" and "rate of return" interchangeably.

A rational investor will not take more risk without expecting a higher return. Since the early 1950s, an enormous amount of theoretical and empirical research has characterized and quantified the relationship between risk and return.

Harry Markowitz initially developed *Portfolio Theory* in the early 1950s. This was the first serious theoretical attempt to quantify the relationship between risk and return. Portfolio Theory often is referred to as *Modern Portfolio Theory* or MPT. Portfolio theory characterizes risk as the uncertainty of returns, and uses standard statistical techniques to quantify the relationship between risk and return. These techniques include statistical measures such as *variance* and standard deviation to quantify the uncertainty of returns.

"Uncertainty of returns" includes not only the possibility of loss, but also the possibility of *positive surprises* relative to expectations. Because investors typically are more concerned about negative surprises than positive surprises, some theoretical work considers only negative deviations relative to expectations. However, risk as the uncertainty of returns, including positive as well as negative deviations from expectations, is the main risk model used in investing.

William Sharpe and others extended portfolio theory to develop asset valuation models, such as the Capital Asset Pricing Model, or CAPM. CAPM introduced the idea of combining a risk-free asset with a portfolio of risky assets to construct a complete portfolio that resulted in an efficient trade-off between risk and return. It also gave a more straightforward way to quantify the risk of individual assets, by comparing their returns with the return of a broad market portfolio. CAPM's other important contribution was to clarify the distinction between *systematic risk* inherent in investing in risky assets, which cannot be eliminated, and *unsystematic risk* specific to individual firms, which diversification can eliminate.

Eugene Fama, Kenneth French, and others added work to develop more sophisticated models to evaluate risk and return. Although CAPM uses a single "factor" to characterize risk, Fama and French showed that a multifactor model better explains historical returns. Instead of comparing the returns of an asset or portfolio of assets only to a broad market portfolio, they also compared them to portfolios of small-cap stocks and value stocks (technically small minus big or SmB, and high book value minus low book value or HmL). This is the *Fama-French three-factor model*.

The finance theory in this article outlines some of the theoretical foundations for asset allocation and investing in index mutual funds. It evaluates expected rates of return for various asset classes and compares them to the risk of the asset classes. Investor goals and risk tolerance determine the trade-off between the expected return and risk of the portfolio. The portfolio combines various lower-risk and higher-risk asset classes to achieve an efficient risk-return trade-off.

Determining risk tolerance is a critical step in designing a portfolio. Several approaches can help investors to assess their risk tolerance.

Both shorter-term risk and longer-term risk are characterized by the uncertainty of returns. Shorter-term risk typically is characterized by the uncertainty of shorter-term returns; for example, monthly or annual returns. Longer-term risk is better characterized by the uncertainty of the cumulative portfolio value (terminal wealth) after many years of investing.

Risk as the uncertainty of returns

The rate of return earned on an investment in a 30-day Treasury bill (T-Bill) or short-term Certificate of Deposit (CD) has high certainty. T-Bills and CDs are among the investments referred to collectively as money market securities.[2] A money market fund provides a convenient way for an investor to own money market securities. The short-term return on deposits in federally-insured bank or credit union accounts is also highly certain (the term *cash* often is used to refer to money market securities and money in deposit accounts; Vanguard refers to these types of assets as *short-term reserves*).

The returns earned by T-Bills, CDs and money market funds fall into a relatively narrow range over longer time periods as well.

The top chart in Figure 1 is a histogram showing the frequency distribution of returns for 3-month T-Bills from 1928 through 2011. The *height* of each bar represents the *number of years* in which annual returns have fallen within a given 5% range or "bin" (that is, 0% to 5%, 5% to 10%, etc.). The tallest bar shows that annual returns have been between 0% and 5% in 59 years. Returns have been between 5% and 10% in 22 years (second tallest bar), and between 10% and 15% in three years (shortest bar).

The return on bonds is less certain than the return on money market securities, so bonds are considered riskier than money market securities (cash). The middle chart in Figure 1 shows the frequency distribution of annual returns for 10-Year Treasury Bonds from 1928 through 2011 (the same vertical scale for number of years is used in all charts in Figure 1, and the same 5% bins are used on the horizontal axes). Notice that the range of returns is larger, from about -11% to +33%, and that the number of years in which returns fall within any given 5% bin is generally smaller. In other words, the shape of the return distribution is more "spread out" and flatter. The shape of the return distribution represents the *dispersion of returns*, also referred to as the *deviation of returns*.

The return on stocks is more uncertain than the return on bonds, so stocks are considered riskier than bonds. The bottom chart in Figure 1 shows the frequency distribution of annual returns on the stocks of large US companies (the S&P 500) from 1928 through 2011. Note the much larger dispersion of returns; the shape of the distribution is much flatter and more spread out, with returns ranging from about -44% to +53%, and none of the 5% bins including returns for more than 10 years (by contrast, recall that one 5% bin included the T-bill returns for 59 years).

Standard deviation is a measure commonly used to quantify the deviation (or dispersion) of returns. The standard deviation is listed for each chart in Figure 1. The standard deviation increases as the dispersion of returns increases (that is, as the shape of the distribution becomes flatter and more spread out). Standard deviation will be defined in more detail in subsequent sections.

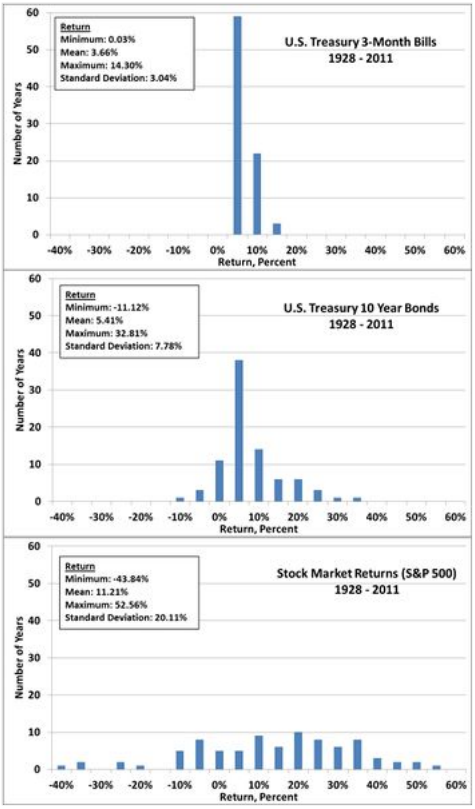


Figure 1. Risk vs. Return: 1928 - 2011[3]

Risk as the possibility of loss

The definition of risk as the uncertainty of investment return does not distinguish between loss and gain. Typically, individual investors think of risk as the possibility that their investments could lose money.[4] They are likely to be quite happy with an investment return that is greater than expected (a "positive surprise"). However, because risky assets generate positive surprises as well as negative surprises, defining risk as the *uncertainty* of the return seems reasonable. Mor uncertainty makes it more likely that the investment will generate gains as well as losses.

Variance and standard deviation, discussed in detail in subsequent sections, quantify the negative *and positive* deviation of returns around the average return (the terms *average return*, *mean return*, and *expected return* are used interchangeably, as are the terms *deviation of returns* and *dispersion of returns*). Some theoretical portfolio work uses measures that consider only risk of loss; semivariance, which considers only deviations below the average, is one example.

However, because deviation of returns is symmetrical, variance provides as good an indicator of downside risk as semivariance. If returns are symmetrical, the probabilities and magnitudes of positive and negative surprises are roughly the same. It is reasonable to assume a symmetrical distribution of returns for a well diversified portfolio, and most assets have returns that are reasonably symmetrical.[5] Therefore in portfolio theory, as well as in practice, variance and standard deviation are the most commonly used measures of dispersion of returns. In other words, even though investors may be primarily concerned with negative surprises, investment risk can be adequately estimated with measures that include both positive and negative surprises.

A major concern of investors is that their investment portfolios will not generate returns sufficient to meet their goals. This is "shortfall risk", and deserves serious consideration.[6] Nevertheless, using uncertainty of returns can be a valid approach in estimating the riskiness of long-term returns as well as short-term returns, as long as it uses the appropriate variance or standard deviation measures.

For long-term returns, it is more appropriate to estimate the *dispersion of accumulated wealth* (also referred to as *terminal wealth dispersion*) rather than the deviation of returns over relatively short time periods, such as annually.[7] This is because relatively small differences in *annualized* returns can make large differences in *cumulative* returns over periods of many years.[8] Long-term dispersion of returns accounts for the *magnitude* as well as the *probability* of possible losses.

"Risk free" assets

Money market securities are often referred to as *risk-free assets*, especially the shorter-maturity securities such as 30-day T-Bills. This is because the short-term return is known with relative certainty at the time the investment is made. There is absolute certainty in the nominal return of a T-Bill (assuming the U.S. government does not default on its obligations), and it is unlikely that unexpected inflation will have significant impact on the real return over a short time period.

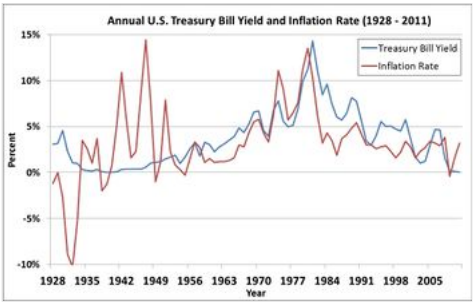


Figure 2. Annualized T-Bill yield and inflation rate (1928 - 2011)[note 1]

Over longer time periods, even money market securities have some risk. This is because the effect of unexpected inflation on returns is uncertain over longer time periods. Although money market security rates usually respond relatively quickly to changes in inflation, this is not always the case. Figure 2 illustrates the longer-term uncertainty of real returns on 90-day T-Bills. The relative *certainty* of return does not mean that the *real* return necessarily is *positive*. It may be known with certainty that a 90-day T-Bill will earn a *nominal* annualized return of 1% over its 90-day term (that is, if held to maturity). However if inflation over the 90-day term is expected to be 3% annualized, the relatively certain, expected *real* return is -2% annualized.

Uncertainty in *real* returns can be eliminated by investing in inflation-indexed securities, such as Treasury Inflation Protected Securities (TIPS) and Series I Savings Bonds (I Bonds), and holding them to maturity. Of course in return for this reduction in uncertainty, investors must accept lower expected returns. Marketable inflation-indexed securities also have other risks, such as interest rate risk (that is, prices decline when interest rates rise) and liquidity risk, as in late 2008 (September 12 - October 31) when the Vanguard Inflation-Protected Securities fund declined by almost 14%. During this same time period other U.S. treasury securities increased in value.[9]

In reality, whether an asset is risk free depends on the context of the investment; that is, the purpose and time horizon of the investment. For example, to meet a nominal liability at a specified future date, a zero-coupon US government bond maturing on that date is essentially risk free.

Nevertheless, short-term T-bills usually are considered to be risk-free assets in portfolio theory, and in practice investors treat a broader range of money market securities as risk free.[10]

Risk aversion and risk premiums

Why would an investor invest in a risky asset class such as stocks rather than in a relatively risk-free asset such as T-bills? If the investor is not simply gambling, there must be some financial incentive, such as the expectation of a higher rate of return. The conclusion is that an investment in a risky asset depends on the investor's expectation of a higher rate of return as well as their level of *risk aversion*.[11]

Rational investors are risk averse. That is, given the same expected return, they will choose the investment for which that return is more certain.[12] Therefore, investors demand a higher expected return for riskier assets. A higher *expected* return does not guarantee a higher *realized* return. Because by definition returns on risky assets are uncertain, an investment may not earn its expected return.

The amount by which a risky asset is expected to provide a higher rate of return than the *risk-free rate* is the asset's *risk premium* (the risk-free rate is the rate of return on a risk-free asset, such as a T-bill). If the risk premium of stocks is zero, a *rational*, risk-averse investor would have no incentive to invest in them. On the other hand, a gambler might "invest" in something with a zero or even negative risk premium for entertainment value, or in the irrational expectation of hitting a jackpot.

Although the charts in Figure 1 show historical (realized) returns rather than expected (future) returns, they are useful to demonstrate the relationship between risk and return, and the risk premiums of bonds and stocks relative to the relatively risk-free returns on T-bills. The mean (average) annual return increases as the dispersion of returns (uncertainty or risk) increases.

Assuming an investor selects an investment portfolio based on expected return and risk, the equation below is used to quantify the relationship between risk premium, risk aversion, A, and risk σ^2, where the risk premium is the expected return, E(r), of the risky investment minus the risk-free rate, r_f:

$$\text{RiskPremium} = \text{E}(r) - r_f = \frac{1}{2} A \sigma^2$$

In this equation risk is measured by variance, σ^2, which quantifies the dispersion of an investment's returns. Variance is discussed in depth in another section. The factor of $^1/_2$ is a scaling factor used by convention, and the use of this equation requires expressing all returns as decimals rather than percentages.

This equation indicates that the risk premium demanded by investors will be higher for individual investors with greater risk aversion, A, and for investments with higher risk, σ^2.

Studies have shown that investors' risk aversion probably is in the range of 2-4.[13] This implies that for an increase of portfolio variance of 0.01, investors will require a risk premium that is higher by 0.01-0.02 (1%-2%).

Measuring risk: historical returns

Historical returns often are used as a starting point in estimating expected returns and uncertainty of returns (risk).[14] Although evaluating historical returns may not enable an accurate or comprehensive estimation of risk, evaluating the dispersion of historical returns provides a convenient way to develop an understanding of commonly used risk measures.

The risk of an investment is related to the uncertainty of its investment return. This was illustrated graphically in Figure 1, which showed that dispersion of returns is a way to characterize the riskiness of an investment. Many investors like to summarize return and risk with numbers that quantify the average return and the dispersion of returns around the average.[15] This makes it easier to compare the return and risk of different investments.[16]

Calculating the average of a set of historical annual returns is straightforward: simply add the annual returns and divide by the number of annual returns. The technical term for this type of average is the *arithmetic mean*, usually referred to simply as the *mean*. Statisticians use the term *expected value* to refer to the mean. In finance theory the term *expected value of the return*, or just *expected return* is commonly used in referring to the average of a set of returns.[17]

What about calculating a number to quantify the uncertainty or dispersion of returns? Basic statistics and probability theory provide two measures to quantify the dispersion of a set of numbers: *variance* and *standard deviation*. Starting in the early 1950s, finance academics began using these statistical measures to quantify risk.[18]

Historical returns are commonly used as a starting point in evaluating the relative riskiness of different assets. A subset of the historical data shown graphically in Figure 1 provides a simple example to begin developing an understanding of standard deviation as a measure of risk.

Consider the S&P 500 stock returns for 2008-2011:

- 2008: -36.55%
- 2009: +25.94%
- 2010: +14.82%
- 2011: +2.07%

The mean, or expected value, E(r), of this set of returns is calculated below (for simplicity, the percent signs are dropped):

$$\text{E}(r) = \frac{-36.55 + 25.94 + 14.82 + 2.07}{4} = 1.57$$

The expected value of the annual return is 1.57%.

An alternate way to calculate expected value is to multiply each value by the probability of that value occurring, then sum the results. This formulation will be useful when calculating the expected value of estimated future returns. When evaluating historical returns, each return is considered to have equal probability, so the calculation using this method is:

$$\text{E}(r) = \frac{1}{4}(-36.55) + \frac{1}{4}(25.94) + \frac{1}{4}(14.82) + \frac{1}{4}(2.07) = 1.57$$

The average of the annual returns is useful information, but it does not indicate anything about the dispersion of returns, that is, how the returns are distributed around the mean return.

It is intuitive that a useful measure of dispersion of returns around the average return would involve calculating the differences (deviations) between the individual returns and the average return, and perhaps averaging these deviations. For example the deviation for the 2009 return is 25.94 - 1.57 = 24.37, whereas the deviation for the much smaller 2011 return is only 2.07 - 1.57 = 0.50.

The problem with this is that there will be both positive and negative deviations that will tend to cancel each other out, resulting in a misleading value that understates the dispersion of returns around the mean; in fact, the average of the deviations from the mean will always be zero.[19] This can be verified by calculating the average of the deviations from the mean for all four years in the example. Therefore, this value indicates nothing about dispersion.

This problem is resolved by *squaring* the deviations from the mean (which results in all positive numbers), then calculating the average of the squared deviations. The term *variance* is used to describe the average of the squared deviations. Here is the calculation of the variance of the annual returns using the returns from 2008-2011:

$$\mathrm{Var}(r) = \sigma^2 = \frac{(-36.55 - 1.57)^2 + (25.94 - 1.57)^2 + (14.82 - 1.57)^2 + (2.07 - 1.57)^2}{4} = 556$$

The symbol σ^2 commonly represents variance, so Var(r) = σ^2 = 556.

This formula can be generalized as:

$$\mathrm{Var}(r) = \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(r_i - \mathrm{E}(r))^2$$

(When calculating the variance of a *sample* of the full population of values (as in the example here), statisticians often substitute $^1/_{(n-1)}$ for $^1/_n$ in the above formula. There are arguments for and against this.[20] In this article this substitution will not be made.)

To get a value with the same units as the rate of return (percent), the square root of the variance is calculated, resulting in the *standard deviation* of the annual returns:

$$\mathrm{SD}(r) = \sigma = \sqrt{556} = 23.6$$

The symbol σ commonly represents standard deviation, so SD(r) = σ = 23.6 pp (technically, the units of the standard deviation of returns are percentage points (pp), but standard deviation of returns often is displayed using the % symbol, for example, 23.6%).

The resulting value of 23.6 pp (or 23.6%) for the standard deviation seems somewhat reasonable as an indicator of the dispersion of returns around the mean value of 1.57%, considering that the range of values is between -36.55% and +25.94%.

Standard deviation has some standard statistical interpretations for a large number of values with a normal distribution (the ubiquitous bell curve). For example, about 68% of all values fall within +/- one standard deviation, and about 95% of all values fall within +/- two standard deviations.

There are too few values in our simple example for these statistical attributes to apply, but consider the 84 annual returns for 10-Year treasury bonds represented in the second chart in Figure 1.

First notice that the data somewhat resembles the bell-shape curve characteristic of a normal distribution. The mean value of the 84 annual returns for 10-Year treasury bonds is about 5%, and the standard deviation is about 8% (pp). Two standard deviations is about 2 x 8 = 16, so we expect most of the annual returns to fall within the range of about 5% +/- 16 pp or between about -11% and +21%. From the chart, this looks about right.

For the third chart in Figure 1, representing 84 annual returns for the stocks in the S&P 500, the mean is about 11% and the standard deviation is about 20%, so we expect most values to fall within a range of about 11% +/- 40 pp, or between about -29% and + 51%. Again, this looks about right.

Rather than calculating the standard deviation of annual returns, the annualized standard deviation of *monthly* returns sometimes is used to characterize dispersion of returns. For example, Morningstar calculates the standard deviation of monthly returns over various time periods of one year or more (for example, three years, five years, and so on), then multiplies by the square root of twelve to generate an estimate of the annualized standard deviation.[21]

Expected return

Although in finance theory, *expected return* is a synonym for the average of a set of returns, in investing, *expected return* more often refers to *future* returns than to *historical* returns. From here on this will be the context in which the term is used. In this context, *expected return* can be thought of informally as the return investors expect to receive on an investment. Indeed, the term often is used this way by authors of investment books (as seems to be the case, for example, in *Expected Returns* by Ilmanen.[22]) However, in investment theory *expected return* has a more precise definition.

An investment's expected return, E(r), is calculated as follows:[23][24]

1. Various economic scenarios, s, are defined.
2. Estimates are developed for the probability, p(s), of each scenario occurring and the return, r(s), for that scenario.
3. The probability and return for each scenario are multiplied together: p(s)r(s)
4. The results are summed across all scenarios: p(s_1)r(s_1) + p(s_2)r(s_2) + … + p(s_n)r(s_n)

Or using summation notation:

$$\mathrm{E}(r) = \sum_{i=1}^{n} \mathrm{p}(s_i)\mathrm{r}(s_i)$$

Thus, expected return is the *probability-weighted average of returns across all possible scenarios*.

First consider an investment in a 1-year T-Bill. Since the nominal return is unaffected by economic factors, the nominal return is certain; that is, the probability, p(*s*) = 1, and the return, r(*s*), is the rate of return on the bill. If the rate of return is 1%, the expected nominal return is calculated as:

$$\mathrm{E}(r) = \mathrm{p}(s)\mathrm{r}(s) = (1.0)(0.01) = 0.01 = 1\%$$

As a slightly more complex, but still simple, scenario analysis, consider a one-year investment in a total stock market index fund, with three possible economic scenarios for the year: recession, normal growth, and strong growth. Assume the following estimates for the three scenarios:

- Recession: probability $\mathrm{p}(s_1) = 0.20$, estimated return $\mathrm{r}(s_1) = -10\%$
- Normal growth: probability $\mathrm{p}(s_2) = 0.60$, estimated return $\mathrm{r}(s_2) = 5\%$
- Strong growth: probability $\mathrm{p}(s_3) = 0.20$, estimated return $\mathrm{r}(s_3) = 10\%$

With these scenario estimates, the expected return,E(*r*), is calculated as:

$$\mathrm{E}(r) = \mathrm{p}(s_1)\mathrm{r}(s_1) + \mathrm{p}(s_2)\mathrm{r}(s_2) + \mathrm{p}(s_3)\mathrm{r}(s_3)$$

$$\mathrm{E}(r) = (0.20)(-0.10) + (0.60)(0.05) + (0.20)(0.10) = 0.03 = 3\%$$

Notice that the expected return of 3% is not equal to the most probable return of 5%.

Extensive research has demonstrated that there are no good forecasters.[25] It follows that scenario analysis itself is fraught with uncertainty. To put it bluntly, expected returns are not directly observable; they can only be estimated.[26] Nevertheless, financial academics and practitioners develop and publish their estimates of expected returns.[27][note 2][28]

A concept that will be useful in subsequent sections is the *expected return of a portfolio*, which is simply the weighted average of the expected returns for the individual assets in the portfolio. This is expressed mathematically as:

$$\mathrm{E}(r_p) = \sum_{i=1}^{n} (w_i)\mathrm{E}(r_i)$$

where w_i is the percent of the portfolio in asset *i*, and $\mathrm{E}(r_i)$ is the expected return of asset *i*.

Measuring risk: estimated returns

A previous section illustrated *variance* of returns using *historical* returns. The calculation of the variance of estimated *future* returns is similar, except that the probabilities of the returns are not equal, but are the probabilities assigned to various scenarios.

Variance of estimated future returns, Var(r), is calculated as follows:

1. Subtract the expected return, E(r), from the estimated return, r(s), for the scenario: [r(s) - E(r)]
2. Square the result: $[\mathrm{r}(s) - \mathrm{E}(r)]^2$
3. Multiply the result by the probability, p(s), of the scenario: $\mathrm{p}(s) \times [\mathrm{r}(s) - \mathrm{E}(r)]^2$
4. Sum the resulting values across all scenarios:

$$\mathrm{Var}(r) = \sigma^2 = \mathrm{p}(s_1)[\mathrm{r}(s_1) - \mathrm{E}(r)]^2 + \mathrm{p}(s_2)[\mathrm{r}(s_2) - \mathrm{E}(r)]^2 + \cdots + \mathrm{p}(s_n)[\mathrm{r}(s_n) - \mathrm{E}(r)]^2$$

Or using summation notation:

$$\mathrm{Var}(r) = \sigma^2 = \sum_{i=1}^{n} \mathrm{p}(s_i)(r_i - \mathrm{E}(r))^2$$

To deepen the intuitive understanding of the calculation of variance of estimated returns, consider the purpose of each step in the calculation.

1. Step 1 yields a number that is larger for a scenario return that is further from the expected return.
2. Step 2 eliminates negative values, so that negative values do not offset positive values; that is, the measure should increase whether a scenario return is less than or greater than the expected return.
3. Step 3 gives a greater weight to a higher-probability scenario.
4. Step 4 adds the individual, probability-weighted, squared differences to yield a single number.

The purposes of steps 1, 2 and 4 are basically the same as for the calculation of the variance of *historical* returns. Step 3 introduces *probability weighting*.

As with historical returns, the *standard deviation* of estimated returns is calculated as the *square root of the variance* of the estimated returns.

For the case of the nominal return on a 1-year T-Bill yielding 1%, the expected return, E(r), = 1%, and there is only one scenario, for which the nominal return is a certain (p(s) = 1). Therefore, the variance is calculated as:

$$\mathrm{Var}(r) = \sigma^2 = \mathrm{p}(s_1)[\mathrm{r}(s_1) - \mathrm{E}(r)]^2$$

$$\mathrm{Var}(r) = \sigma^2 = (1.0)(0.01 - 0.01)^2$$

$$\mathrm{Var}(r) = \sigma^2 = 0$$

Since √0 = 0, the standard deviation is 0 pp. Note the intuitive sense of this: a standard deviation of 0 indicates zero uncertainty in the return.

Recall the three-scenario case discussed earlier:

- Recession: probability, $p(s_1) = 0.20$; estimated return, $r(s_1) = -10\%$
- Normal growth: probability, $p(s_2) = 0.60$; estimated return, $r(s_2) = 5\%$
- Strong growth: probability, $p(s_3) = 0.20$; estimated return, $r(s_3) = 10\%$

The expected return, E(r), was calculated as 3%. The variance is calculated as:

$$\mathrm{Var}(r) = \sigma^2 = \mathrm{p}(s_1)[\mathrm{r}(s_1) - \mathrm{E}(r)]^2 + \mathrm{p}(s_2)[\mathrm{r}(s_2) - \mathrm{E}(r)]^2 + \mathrm{p}(s_3)[\mathrm{r}(s_3) - \mathrm{E}(r)]^2$$

$$\mathrm{Var}(r) = \sigma^2 = (0.20)(-0.10 - 0.03)^2 + (0.60)(0.05 - 0.03)^2 + (0.20)(0.10 - 0.03)^2$$

$$\mathrm{Var}(r) = \sigma^2 = (0.20)(-0.13)^2 + (0.60)(0.02)^2 + (0.20)(0.07)^2$$

$$\mathrm{Var}(r) = \sigma^2 = 0.0046$$

And the standard deviation is:

$$\mathrm{SD}(r) = \sigma = \sqrt{0.0046}$$

$$\mathrm{SD}(r) = \sigma = 0.07 = 7\%$$

Although three samples is not enough to apply the statistical implications of standard deviation, two of the returns fall within E(r) +/- one standard deviation (3% +/- 7 pp), and all returns fall within E(r) +/- two standard deviations (3% +/- 14 pp). Again, standard deviation seems to provide a reasonable measure of the dispersion of returns (risk).

Managing portfolio risk

A *portfolio* is the investor's collection of financial assets, such as stocks, bonds and cash. Selecting investments to balance the tradeoff between risk and expected return is *portfolio selection*, *portfolio construction*, or *portfolio optimization*. *Portfolio Theory* is the theoretical basis for portfolio selection.

Portfolio selection has two main techniques for managing the risk-return tradeoff: diversification, and asset allocation. Diversification means owning enough securities in different industries to reduce or eliminate firm-specific and industry-specific risks. Asset allocation involves building an efficient portfolio, using multiple asset classes (such as stocks, bonds, and cash), that gives an appropriate risk-return tradeoff based on the investor's risk tolerance.

Intuitively, portfolio risk is proportional to the amount of risky assets in the portfolio. A straightforward way to adjust the risk of the complete portfolio is to combine a portfolio of risky assets with a risk-free asset, for example combining a total stock market index fund (the risky portfolio) with T-bills or other money market securities (the risk-free asset). The risk of the complete portfolio will be proportional to the proportion of the risky portfolio included in the complete portfolio, and inversely proportional to the proportion of the risk-free asset. Of course portfolio theory posits that expected return also is proportional to portfolio risk.

Although less intuitive, investors can adjust the portfolio risk-return tradeoff by combining different risky assets in various proportions. For example, they can improve the risk-return tradeoff by selecting a risky portfolio that includes US stocks, non-US stocks, and bonds. A goal of portfolio selection is to find a combination of risky assets that optimizes the ratio of expected return to risk. The ratio of expected return to risk is considered a measure of *portfolio efficiency*.

Classic portfolio selection consists of two steps:

1. Determine an optimal combination of risky assets (the *risky portfolio*).
2. Construct the *complete portfolio* by combining the risky portfolio with a risk-free asset in proportions that achieve an appropriate ratio of expected return to risk, based on the investor's risk tolerance.

The resulting portfolio is an efficient portfolio, in that any other combination of risky and risk-free assets would have either a lower expected return for a given level of risk, or more risk for a given level of expected return. Of course since expected returns and risk are not observable, but can only be estimated, portfolio efficiency cannot be known with any great certainty. A portfolio optimized for historical returns is unlikely to remain efficient. Nevertheless, investors often use historical returns to help estimate appropriate proportions of different risky asset classes to include in a portfolio.

Because step 2 of the two-step process described above involves simpler theory, it will be covered first.

Portfolio of risky assets combined with a risk-free asset

Risky assets include bonds as well as stocks, but for now assume that the risky portfolio is a total stock market index fund. As discussed in previous sections, there is no truly risk-free asset, but T-bills often are considered the risk-free asset in portfolio theory. The risk of T-bills and other money market securities is so much lower than the risk of stocks that this is a reasonable approach, especially for relatively short holding periods.

Both the *expected return* and the *risk* of a portfolio must be determined to evaluate the *risk-return trade-off* of combining a portfolio of risky assets with a risk free asset. The following steps develop an equation that relates the expected return of a such a portfolio to its risk, where risk is measured by the standard deviation of portfolio returns.

Portfolio of risky assets combined with a risk-free asset: Expected return

The expected return of a portfolio of assets is the weighted average of the expected returns of the individual assets:

$$\mathrm{E}(r_p) = w_1\mathrm{E}(r_1) + w_2\mathrm{E}(r_2) + \cdots + w_n\mathrm{E}(r_n).$$

This can be expressed more concisely using summation notation as:

$$\mathrm{E}(r_p) = \sum_{i=1}^{n} w_i \mathrm{E}(r_i)\,.$$

The *weight* of an asset in a portfolio refers to the fraction of the portfolio invested in that asset. For example, if w_1 = ¼, then one fourth of the portfolio is invested in asset *1* with expected return E(r_1).

For *two assets* this reduces to:

$$\mathrm{E}(r_p) = w_1 \mathrm{E}(r_1) + w_2 \mathrm{E}(r_2)$$
$$\mathrm{E}(r_p) = w_1 \mathrm{E}(r_1) + (1 - w_1) \mathrm{E}(r_2)$$

Let one asset be the risky portfolio consisting of a total stock market index fund, with expected return $\mathrm{E}(r_s)$ = 6%, and with the standard deviation of annual returns = 20% (these values are very close to the values for the historical returns of the Vanguard Total Stock Market Index fund from 1998 through 2011). Let the other asset be a risk-free asset with return r_f = 1% (since r_f is known with certainty, $\mathrm{E}(r_f) = r_f$). The rate of return of the risk-free asset is referred to as the *risk-free rate of return*, or simply the *risk-free rate*. The standard deviation of the risk-free asset is 0% by definition. Applying the above equation to this portfolio:

$$\mathrm{E}(r_p) = w_s \mathrm{E}(r_s) + (1 - w_s)(r_f)$$
$$\mathrm{E}(r_p) = w_s \mathrm{E}(r_s) - w_s r_f + r_f$$
$$\mathrm{E}(r_p) = r_f + w_s(\mathrm{E}(r_s) - r_f)$$

$\mathrm{E}(r_s) - r_f$ is the *risk premium* of the risky portfolio. The expected risk premium of an asset is the expected return of the asset above the risk-free rate. Since the risky portfolio here is a stock fund, its risk premium is referred to as the *equity risk premium* or ERP (*equities* is synonymous with *stocks*).

Plugging in the risk-free rate and the expected return for the stock fund gives:

$$\mathrm{E}(r_p) = 0.01 + w_s(0.06 - 0.01)$$
$$\mathrm{E}(r_p) = 0.01 + 0.05 w_s$$

This is a linear equation indicating that a portfolio of any expected return between r_f = 1% and $\mathrm{E}(r_s)$ = 6% can be constructed by combining the risky portfolio and risk-free asset in the desired proportions. The risk premium of the stock fund is 0.05 = 5%.

If w_s = 0, the portfolio consists only of the risk-free asset, and the expected return of the portfolio is simply the risk-free rate:

$$\mathrm{E}(r_p) = 0.01 + (0.05)(0)$$
$$\mathrm{E}(r_p) = 0.01 = 1\%$$

If w_s = 1, the total portfolio consists entirely of the risky portfolio, and the expected return of the total portfolio is the expected return of the risky portfolio:

$$\mathrm{E}(r_p) = 0.01 + (0.05)(1)$$
$$\mathrm{E}(r_p) = 0.06 = 6\%$$

If the portfolio is divided evenly between the risky portfolio and the risk-free asset, w_s = 0.5, and the expected return of the portfolio is:

$$\mathrm{E}(r_p) = 0.01 + (0.05)(0.5)$$
$$\mathrm{E}(r_p) = 0.01 + (0.025)$$
$$\mathrm{E}(r_p) = 0.035 = 3.5\%$$

In the case of a portfolio evenly divided between the risky portfolio and the risk-free asset, $w_s = w_f$ = ½, and the expected return is simply the average of the returns of the two assets:

$$\mathrm{E}(r_p) = \frac{0.01 + 0.06}{2} = 0.035 = 3.5\%$$

Portfolio of risky assets combined with a risk-free asset: Risk

Because the standard deviation of a risk-free asset is 0%, the standard deviation of a portfolio formed by combining a portfolio of risky assets with a risk-free asset is simply the standard deviation of the risky portfolio times the proportional weight of the risky portfolio:

$$\mathrm{SD}(p) = \sigma_p = w_s \sigma_s$$

So the risk of the total portfolio is linearly proportional to the weight of the risky portfolio. Given the standard deviation of annual returns of the risky portfolio of stocks = 20%, a portfolio can be constructed with a standard deviation between 0% and 20%. A portfolio evenly divided between the risky portfolio and the risk-free asset would have a standard deviation of 10%.

Portfolio of risky assets combined with a risk-free asset: expected return in terms of risk

The above equations can be combined to express the expected return of the total portfolio as a function of its standard deviation, thus *relating the expected return of the portfolio to its risk*. Rearranging the equation for the standard deviation of the total portfolio gives $w_s = \sigma_p / \sigma_s$. Therefore,

$$\mathrm{E}(r_p) = r_f + w_s[\mathrm{E}(r_s) - r_f]$$
$$\mathrm{E}(r_p) = r_f + \frac{\sigma_p}{\sigma_s}[\mathrm{E}(r_s) - r_f]$$

$$\mathrm{E}(r_p) = r_f + \frac{[\mathrm{E}(r_s) - r_f]}{\sigma_s}\sigma_p$$

This is a linear equation that gives portfolio expected return, $\mathrm{E}(r_p)$, as a function of portfolio risk as measured by standard deviation, σ_p, with intercept r_f and slope $\frac{\mathrm{E}(r_s) - r_f}{\sigma_s}$. The line plotted from this equation is known is the *Capital Allocation Line* or *CAL*, since capital can be allocated between the risky portfolio and risk-free asset to achieve any risk-return combination along the line.

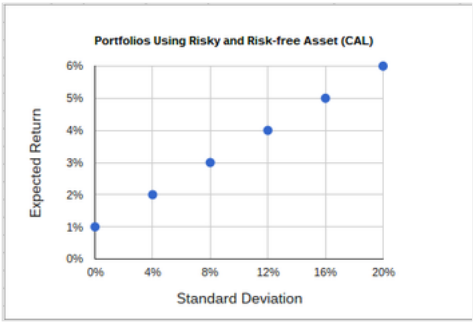
The slope of the CAL is the ratio of the risky portfolio's excess expected return (risk premium) to its risk (that is, its variability as measured by standard deviation). This reward-to-variability ratio is a widely used measure of portfolio efficiency (risk-adjusted return), and is commonly referred to as the *Sharpe ratio* (or Sharpe measure), after William Sharpe who first suggested its use.[29] The steeper the slope of the CAL, the more efficient the portfolios to choose from.

Although the theory being discussed here involves expected (ex-ante) returns, the Sharpe ratio also is commonly used to evaluate the efficiency or risk-adjusted returns of portfolios (for example, mutual funds) based on historical (ex-post) returns.

Substituting in the example numbers of $r_f = 0.01$, $E(r_s) = 0.06$, and $\sigma_s = 20\% = 0.20$:

$$\mathrm{E}(r_p) = r_f + \frac{[\mathrm{E}(r_s) - r_f]}{\sigma_s}\sigma_p$$
$$\mathrm{E}(r_p) = 0.01 + \frac{0.06 - 0.01}{0.20}\sigma_p$$
$$\mathrm{E}(r_p) = 0.01 + 0.25\sigma_p$$

So a portfolio of any expected return between 1% and 6% with proportional standard deviation between 0% and 20% can be constructed by combining the risk-free asset and the risky portfolio in appropriate proportions. With a CAL slope of 0.25, portfolio expected return increases by 1 percentage point for each increase of 4 percentage points in portfolio standard deviation (σ_p); that is, solving this equation for values of σ_p = 0%, 4%, 8%, 12%, 16% and 20% gives values for $E(r_p)$ of 1%, 2%, 3%, 4%, 5% and 6%, as shown in the chart in the nearby figure.



Investment Opportunity Set of Portfolios Using a Risk-Free Asset and One Risky Asset (Capital Allocation Line)

Portfolio of two risky assets

This section develops the theory that shows the risk-return characteristics for a portfolio consisting of two risky assets. It will be extended to more than two risky assets in the following section. This is the theory that underlies the practice of combining multiple risky assets into a portfolio that has less risk than the weighted sum of the risks of the individual assets. Examples are a mutual fund that includes many individual securities, and a portfolio of mutual funds that includes different asset classes; for example, domestic stocks, foreign stocks, and bonds.

In considering two risky assets, it is convenient to consider two mutual funds, but the same theory applies to any two assets; for example, two individual stocks. In the following section in which portfolios of many assets are considered, it will be more convenient to consider individual securities.

The statistical measures *covariance* and *correlation* are fundamental to calculating the risk-return characteristics of a portfolio consisting of multiple risky assets.

Covariance is a measure of how the values of two dependent variables move together, or co-vary, as the value of a common independent variable changes.[30] The dependent variables of interest in portfolio theory are the returns of the assets in the portfolio, and the independent variable could be a scenario (for example, the state of the economy) or a time period.

Consider the product of the deviations of returns for two assets, stock fund *s* and bond fund *b*, from their expected values for a particular scenario (or during a certain time period):

$$[r_s - \mathrm{E}(r_s)][r_b - \mathrm{E}(r_b)]$$

If the returns of the two assets are both above their expected values (both deviations are positive), or the returns of the two assets are both below their expected values (both deviations are negative), the product of the deviations will be positive. If the return of one asset is above its expected value (positive deviation), and the return of the other asset is below its expected value (negative deviation), the product of the deviations will be negative. Thus the sign of the product of the deviations indicates whether the returns have deviated in the same direction or opposite directions, and the magnitude of the product of the deviations indicates how much the returns have deviated from their expected values. The covariance of returns is the expected value of the products of the deviations across all scenarios:

$$\mathrm{Cov}(r_s, r_b) = \sum_{i=1}^{n} p(i)[(r_s(i) - \mathrm{E}(r_s)][r_b(i) - \mathrm{E}(r_b)]$$

where p(i) is the probability for scenario i, and $r_s(i)$ and $r_b(i)$ are the respective returns for the stock and bond fund for scenario i.

Often a more convenient measure to work with is the *correlation coefficient*, which is computed by dividing the covariance by the product of the standard deviations of the two assets:

$$\rho_{s,b} = \frac{\mathrm{Cov}(r_s, r_b)}{\sigma_s \sigma_b}$$

The correlation coefficient varies from -1 for two perfectly uncorrelated variables to 1 for perfectly correlated variables. A correlation coefficient of 0 indicates no correlation between the two variables.

The variance of the rate of return of a portfolio consisting of two risky assets, in this case stock fund *s* and bond fund *b*, is given by this equation:

$$\sigma_p^2 = (w_s\sigma_s)^2 + (w_b\sigma_b)^2 + 2(w_s\sigma_s)(w_b\sigma_b)\rho_{s,b}$$

or equivalently

$$\sigma_p^2 = (w_s\sigma_s)^2 + (w_b\sigma_b)^2 + 2w_s w_b \mathrm{Cov}(r_s, r_b)$$

where w_s and w_b represent the respective proportions (weights) of the portfolio invested in the stock and bond funds, and σ_s and σ_b are the respective standard deviations of the stock and bond funds.

The portfolio variance depends on the weighted standard deviations of the assets *plus* a term that depends on the covariance (or correlation) of the assets. The smaller the correlation between the assets the lower the portfolio variance.

For a correlation coefficient of 1 (perfectly correlated assets), the equation for portfolio variance is:

$$\sigma_p^2 = (w_s\sigma_s)^2 + (w_b\sigma_b)^2 + 2(w_s\sigma_s)(w_b\sigma_b)$$

which reduces to:

$$\sigma_p^2 = (w_s\sigma_s + w_b\sigma_b)^2$$

and the standard deviation is:

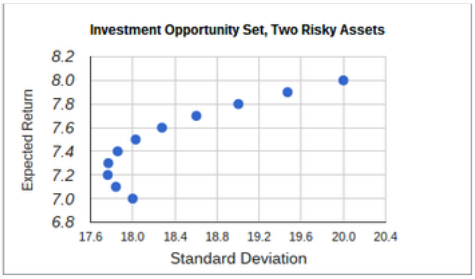
$$\mathrm{SD}(p) = \sigma_p = (w_s\sigma_s + w_b\sigma_b)$$

So for two perfectly, positively correlated assets, the portfolio standard deviation is simply the weighted average of the standard deviations of the two assets. *Any correlation less than 1 will result in a portfolio standard deviation that is less than the weighted average of the standard deviations of the individual assets.*

Since the portfolio expected return is the weighted average of the returns of the individual assets (as shown in previous sections), but the standard deviation is less than the weighted average of the assets' standard deviations, *a portfolio of assets that are not perfectly correlated always provides a better risk-return opportunity than the individual assets on their own.*[31]

As a simple illustration of this important concept, consider a portfolio of two assets that have the same expected value and standard deviation, but that are not perfectly correlated (correlation coefficient less than 1). The expected value of the portfolio will be the same as the expected value of each asset, but the variance (and standard deviation) of the portfolio will be less than the variance (and standard deviation) of each asset (this can be verified with the above equations). This demonstrates how combining risky assets can result in a portfolio that is less risky than either asset on its own.

To graphically illustrate the diversification benefit of a portfolio consisting of two risky, imperfectly correlated assets, assume estimates of expected returns of 7% and 8% respectively for a total US stock fund and a total international stock fund, respective standard deviations of 18% and 20%, and a correlation coefficient of 0.8. The nearby figure shows a graph of expected return vs. standard deviation for portfolios consisting of various proportions of each fund.



Investment Opportunity Set of Portfolios Formed by Combining Two Risky Assets in Various Proportions

The lower left point represents the return and standard deviation of the US stock fund, and the upper right point represents return and standard deviation of the international stock fund. Every other point represents a portfolio consisting of both funds. Each point represents a 10 percentage point change in the proportions of each fund. The expected return and standard deviations for each point are calculated using the above equations.

Notice that a portfolio consisting of 60% US stocks and 40% international stocks provides a higher expected return (about 7.4%) with a lower standard deviation (about 17.9%) than a portfolio consisting of 100% US stocks. This is because the correlation coefficient, 0.8, is less than 1. If the correlation coefficient were 1, the investment opportunity set would consist of a straight line between the lower left point and the upper right point. The degree to which the investment opportunity set curve bulges to the left of a straight line connecting the endpoints represents the diversification benefit due to imperfect correlation.

Of course expected returns, standard deviations, and correlation coefficients cannot be known in advance, so the diversification benefit of combining two risky assets is only as good as your estimates of these values. It is common to use historical values as a starting point in estimating the inputs required to construct an efficient portfolio.

For example, based on their analysis of historical returns, Vanguard's research group as concluded that it is reasonable for US investors to allocate between 20% and 40% of the stock portion of their portfolios to international stocks.[32] Vanguard currently allocates 30% to international stocks in the stock portions of their balanced index funds, such as the Target Retirement and LifeStrategy funds.

Portfolio of many risky assets

The general equation for the variance of a portfolio of risky assets is:

$$\sigma_p^2 = \sum_{i=1}^{n} w_i^2\sigma_i^2 + \sum_{i=1}^{n}\sum_{j=1,\, j\neq i}^{n} w_i w_j \sigma_i \sigma_j \rho_{ij},$$

Recalling that $\mathrm{Cov}(i,j) = \sigma_i\sigma_j\rho_{ij}$, the equation can be rewritten more succinctly as:

$$\sigma_p^2 = \sum_{i=1}^{n} w_i^2\sigma_i^2 + \sum_{i=1}^{n}\sum_{j=1 j\neq i}^{n} w_i w_j \mathrm{Cov}(i,j)\,.$$

The terms summed on the left are the *variances of the individual assets* multiplied by the squared portfolio weights of the assets. The double summation on the right represents the summation across all assets of the *covariance of each pair of assets* multiplied by the portfolio weights of the assets in each pair.

The condition $j \neq i$ indicates that the covariance of an asset with itself is not included in the double summation. The covariance of an asset with itself is the asset's variance — that is, $\mathrm{Cov}(i,i) = \sigma_i^2$ — and the assets' variances already are included in the summation on the left. To clarify this point, the equation can be rewritten as:

$$\sigma_p^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \mathrm{Cov}(i,j)\,.$$

When $i = j$, the term included in the sum is:

$$w_i w_i \mathrm{Cov}(i,i) = w_i^2\sigma_i^2,$$

which represents term i in the first summation in the equation as previously written. So factoring out the sum of terms where $i = j$ gives the equation as previously written.

Expanding this equation for a portfolio of two risky assets gives:

$$\sigma_p^2 = (w_1\sigma_1)^2 + (w_2\sigma_2)^2 + w_1 w_2 \mathrm{Cov}(1,2) + w_2 w_1 \mathrm{Cov}(2,1)\,.$$

There are two variance terms and two covariance terms. Since $\mathrm{Cov}(1,2) = \mathrm{Cov}(2,1)$, the equation can be simplified to:

$$\sigma_p^2 = (w_1\sigma_1)^2 + (w_2\sigma_2)^2 + 2w_1 w_2 \mathrm{Cov}(1,2)\,.$$

So although there two covariance terms before simplifying, only one covariance term must be calculated. This is the equation used in the previous section.

For three risky assets the equation expands to:

$$\sigma_p^2 = (w_1\sigma_1)^2 + (w_2\sigma_2)^2 + 2w_1 w_2 \mathrm{Cov}(1,2) + 2w_1 w_3 \mathrm{Cov}(1,3) + 2w_2 w_3 \mathrm{Cov}(2,3)\,.$$

There are three variance terms and six original covariance terms, three of which must be calculated after simplifying the equation as written. As more assets are added to the portfolio, the number of covariance terms becomes much larger than the number of variance terms. Specifically, there are n variance terms and n(n-1) covariance terms (but only n(n-1)/2 covariance terms that must be calculated). It is intuitive that for a large number of assets the covariance terms will dominate the result; that is, *the covariance between asset pairs will have much more impact on overall portfolio variance than the variances of the individual assets*.

For asset pairs with given expected returns, lower covariance (correlation) between asset pairs in the portfolio results in lower portfolio variance and a more efficient portfolio. Portfolio efficiency is the ratio of expected excess portfolio return to portfolio variance. This demonstrates the diversification benefit of building a portfolio of many assets with less than perfect correlation.

To further illustrate the benefit of diversification, consider the case in which a portfolio is constructed of many assets held in equal proportions. In this case $w_i = {}^1/_n$ for each asset, and the equation for portfolio variance becomes:

$$\sigma_p^2 = \sum_{i=1}^{n} (\frac{1}{n})^2\sigma_i^2 + \sum_{i=1}^{n}\sum_{j=1 j\neq i}^{n} (\frac{1}{n})(\frac{1}{n})\mathrm{Cov}(i,j)\,.$$

With appropriate factoring, this equation can be rewritten as:

$$\sigma_p^2 = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{n}\sigma_i^2 + \frac{n-1}{n}\sum_{i=1}^{n}\sum_{j=1 j\neq i}^{n}\frac{\mathrm{Cov}(i,j)}{n(n-1)}.$$

The summation of terms on the left is the average variance of the assets. As n becomes large *1/n* approaches 0, so the term on the left approaches 0. This term represents the *nonsystematic risk* of the individual assets that can be diversified away by including a sufficiently large number of assets in the portfolio. In other words, in a portfolio of many assets, the variances of the individual assets have little impact on the variance of the portfolio.

The double summation of terms on the right is the average covariance of the assets. As n becomes large, *(n-1)/n* approaches 1, so the term on the right approaches the average covariance. This term represents the *systematic risk* of the portfolio that cannot be diversified away.

This further demonstrates that it is the covariance between the assets in a portfolio of many assets that dominates the overall portfolio risk, and that the variances of the individual assets have little impact on the overall portfolio risk.

The sources of systematic risk are the economic factors that affect the overall market. For example, the level of economic growth affects all companies to some extent, and therefore affects the returns of all stocks to some extent.

The portion of an asset's return variation that is *not* related to the market portfolio is referred to as *unsystematic risk* or *nonsystematic risk*. The sources of unsystematic risk are the *firm-specific risks* of individual companies, as opposed to the broader economic factors that have some effect on all companies.

As shown above, unsystematic risks can be eliminated, or "diversified away", by owning a portfolio of many assets that are not perfectly correlated. So one way to manage risk is to reduce or eliminate unsystematic risk by holding a large number of such securities--ultimately by holding all securities in the market. This is the theoretical foundation for owning a total market mutual fund, such as a total stock market or total bond market index fund.

Determining the expected return and variance of a portfolio using the equations developed so far requires estimating *n* expected returns, *n* variances, and *n(n-1)/2* covariances. For a portfolio of many assets, the number of covariances that must be estimated becomes problematic. Index models provide a solution to this problem. An index model is an implementation of a factor model. The Capital Asset Pricing Model, or CAPM, provides the theoretical foundation for a single-factor model.

CAPM and single factor models

Given a simplifying set of assumptions, CAPM provides an equation that relates the excess expected return of asset *i* to its expected covariance with the expected excess return of the market portfolio *M*:

$$\mathrm{E}(r_i) - r_f = \frac{\mathrm{Cov}(i,M)}{\sigma_M^2}(r_M - r_f).$$

Excess expected return is the expected return of the asset or market portfolio minus the risk-free rate r_f. The market portfolio is a theoretical portfolio consisting of all marketable, risky assets; for example, stocks, bonds, real estate, commodities, collectibles, and so on.

Isolating the expected return of asset *i* on the left hand side of the equation gives:

$$\mathrm{E}(r_i) = r_f + \frac{\mathrm{Cov}(i,M)}{\sigma_M^2}(r_M - r_f).$$

Defining the term $\frac{\mathrm{Cov}(i,M)}{\sigma_M^2}$ as beta (β_i), the equation can be rewritten as:

$$\mathrm{E}(r_i) = r_f + \beta_i(r_M - r_f).$$

Beta standardizes the measure of systematic risk by dividing the covariance of the asset with the market portfolio, $\mathrm{Cov}(i,M)$, by the variance of the market portfolio, σ_M^2. Since the covariance of an asset with itself is the variance of the asset, this results in a beta of 1 for the market portfolio. The *expected return-beta* equation above is the formulation of CAPM most widely used by investment professionals.[33]

The CAPM model relates expected return to expected risk, and involves a theoretical market portfolio consisting of all risky assets. Since expected returns are not observable, and in practice the market portfolio is not observable, factor models and index models were developed to enable investment professionals to put the theory behind CAPM into practice.

In a *single-factor model* it is assumed that all unexpected macroeconomic events affecting the asset returns in a market can be summarized by one common factor, and that all remaining uncertainty in asset returns is firm specific.[34] Recognizing that each firm will have a different sensitivity to macroeconomic events, the following equation relates the return on security *i* to the common macroeconomic factor *F*:

$$r_i = \mathrm{E}(r_i) + \beta_i F + e_i,$$

where $\mathrm{E}(r_i)$ is the expected return on security *i* at the beginning of the holding period, β_i is the sensitivity of security *i* to macroeconomic factor *F*, and e_i is component of return due to the impact of unanticipated firm-specific events.

A *single-index model* uses the excess return on a market index, such as the S&P 500, as a proxy for the common macroeconomic factor. Broader proxies for the market portfolio can be used; for example, all US stocks, all global stocks, or all global stocks and bonds. A single-index model equation for the excess holding period return on security *i* is:

$$r_i - r_f = \alpha_i + \beta_i(r_M - r_f) + e_i\,,$$

where α_i is the security's expected return if the excess market return is 0%, r_M is the market index return, β_i is the sensitivity of security *i* to the market index, and e_i is the component of return due to unexpected firm-specific events.

Some authors express the single-index model in terms of total returns rather than excess returns.[35]

Using capital *R* to represent excess return, the equation can be written more simply as:

$$R_i = \alpha_i + \beta_i R_M + e_i\,.$$

In this equation $\beta_i R_M$ represents a source of *systematic risk* based on the security's sensitivity to the market index (a proxy for macroeconomic factors), and e_i represents a source of *unsystematic risk* based on firm-specific events.

Since expected returns are not observable, beta typically is calculated using periodic historical returns over a specified time period; for example, weekly or monthly returns over the previous three to five years. For example, Morningstar calculates beta based on monthly returns over the previous 36 months.[36] The above equation is used as the basis of a linear regression analysis in which the dependent variable is the excess return of the individual asset, R_i, the independent variable is the excess return of the market index, R_M, and beta is the regression coefficient (slope of the line).

This can be represented graphically as follows. A scatter plot of R_i vs. R_M is generated in which the vertical axis measures the excess returns of the individual asset, and the horizontal axis measures the excess returns of the market index. Each point in the scatter plot represents a pair of excess returns for one period (week or month). A best-fit line is drawn through the points. The slope of the line is beta, and the vertical intercept is alpha. The vertical distance from the line of each point for period *i* is e_i, the deviation from the regression line.

The variance of the excess return on security *i* can be written as:

$$\mathrm{Var}(R_i) = \sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma^2(e_i).$$

On the right hand side of the equation, the first term is the variance due to uncertainty in the common macroeconomic factors (as represented by the market index), and the second term is the variance due to firm-specific uncertainty.

The covariance between the excess rates of return on any two securities is:

$$\mathrm{Cov}(R_i, R_j) = \mathrm{Cov}(\beta_i R_M, \beta_j R_M) = \beta_i \beta_j \sigma_M^2.$$

From these equations it can be determined that $(3n + 1)$ estimates are required to evaluate the portfolio variance for n securities. This is many fewer estimates than the $2n + n(n - 1)/2$ estimates required using variances and covariances directly.

Multifactor models

Although CAPM and the related single-factor model have had significant impact on finance theory and practice, empirical studies have shown significant discrepancies between historical data and predictions of the CAPM-based single factor model. This may not be surprising considering the simplification of assuming that only a single factor affects the systematic variation in security prices. Theoretical work has been done attempting to use more than one factor to explain the systematic risk of securities.

Two of the discrepancies mentioned above are the long-term returns of small company stocks and stocks with high ratios of book value to market value (value stocks), which are higher than predicted by CAPM. Motivated by these anomalies, Fama and French conducted research that resulted in what is commonly referred to as the Fama-French three-factor model.

One of the three factors is the market portfolio (more specifically, an index representing the market), as in CAPM. The other two factors are SmB (Small minus Big) and HmL (High minus Low). SmB is a theoretical portfolio represented by the returns of small-cap stocks minus the returns of large-cap stocks (hence, Small minus Big or SmB). HmL is a theoretical portfolio represented by the returns of high book-to-market stocks minus the returns of low book-to-market stocks (hence, High minus Low or HmL).

Fama and French's research indicated that regressing the empirical data against the three factors explained the returns of stock portfolios much better than did the CAPM-based single-factor model. The equation used to perform the regressions is:

$$r_i - r_f = \alpha_i + \beta_M (r_M - r_f) + \beta_h r_{HmL} + \beta_s r_{SmB} + e_i \,.$$

Each of the betas in the equation represent the sensitivity of a stock or portfolio of stocks to the corresponding factor. The empirically derived beta values can be considered as explaining how much of a portfolio's return can be explained by exposure to the respective factors. A portfolio more heavily weighted toward small-cap and value stocks would be expected to have higher values of β_s and β_h.

This result is based on empirical research, not a theory related to specific risk factors. Therefore, one conclusion is that SmB and HmL may be *proxies* for exposure to sources of systematic risk not captured by the single CAPM factor of overall market risk.

Investors who believe that SmB and HmL are valid proxies for systematic risk factors may decide to increase their allocations to small-cap stocks and value stocks. This is informally referred to as *tilting* toward small and value.

Assessing risk tolerance

Main article: Assessing risk tolerance

Assessing risk tolerance is a critical step in determining the appropriate trade off between the risk and expected return of a portfolio. Selecting a mix of risky and risk-free assets (or higher-risk and lower-risk assets) is one of the most important decisions in designing a portfolio. The investor must develop a rational assessment of risk tolerance to make rational decisions about this asset allocation during portfolio design.

Criteria

Most approaches to assessing risk tolerance consider attitude toward risk and investment time horizon in one way or another. Other factors that affect ability to compensate for investment losses include net worth, stability of income, future liabilities, and flexibility of goals.

- The Bogleheads' Guide To Investing explores four areas: Goals, Time frame, Risk tolerance, Personal financial situation. Tools are recommended to help factor risk into your asset allocation. Various sample portfolios are presented based on these "stages in life".
- Bogleheads author Larry Swedroe suggests that investors evaluate their risk tolerance by considering their *ability, willingness and need* to take risk.
- Bogleheads author William Bernstein discusses determining appropriate risk exposure in the context of deciding on your allocation between stocks and bonds based on age and risk tolerance (attitude toward risk), at least as a starting point.
- Bogleheads author Rick Ferri suggests using a modified version of the age in bonds rule of thumb to help decide on the split between higher-risk and lower-risk assets, and presents asset allocation decisions in terms of a life-cycle investing framework.
- Daniel R. Solin in his book *The Smartest Money Book You'll Ever Read*, has a full chapter titles *Assessing Your Risk Capacity* where he presents five major factors for assessing risk capacity.
- The CFA Institute discusses Investors and Objectives, Investor constraints (Liquidity, Investment Horizon, Regulations, Tax Considerations, Unique Needs.

Self-assessment questionnaires

Various advisory services offer self-assessment questionnaires to help determine your risk tolerance and define an initial asset allocation. Such tools are useful with caveats.

Managing financial risk

Investment risk, or financial-asset risk, is only one type of financial risk facing households. Other major categories of risk are:[37]

- Sickness, disability, and death.
- Unemployment risk.
- Consumer-durable asset risk: for example, the risk of loss from fire, theft or obsolescence related to owning a car, house, or other consumer-durable asset.
- Liability risk: for example, the possibility of being sued.

Managing these types of risks, as well as managing investment risk, from a broader financial planning perspective is the topic of this section.

Short term risk vs. long term risk

Mentioned in "Risk as the possibility of loss" section, perhaps here we go into more detail on terminal wealth dispersion, how small differences in annualized return can result in large differences in terminal wealth, stocks *are* risky in the long run, and so on. This is covered in Bodie, Kane, Marcus, 2008.

See Bernstein's "The 15 Stock Myth" (https://www.efficientfrontier.com/ef/900/15st.htm) for more on terminal wealth dispersion, but there he focuses on the danger of a portfolio with only a small number of stocks. He does emphasize that such portfolios, although they may have low standard deviation of monthly returns, also have a large dispersion of terminal wealth.

Notes

1. T-bill data source comes from Annual Returns on Stock, T.Bonds and T.Bills: 1928 - Current (https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html), Damodaran Online. Inflation data comes from Consumer Price Index 1913 -, Minnesota Federal Reserve (https://www.minneapolisfed.org/community/teaching-aids/cpi-calculator-information/consumer-price-index-and-inflation-rates-1913).
2. Various links to GMO's latest 7-year forecast can be found by searching for "GMO 7-Year Asset Class Forecasts (February 2012)". The latest GMO *7-Year Asset Class Return Forecasts* is available directly on the GMO website (https://www.gmo.com/America/Research/), but registration (free) is required.

See also

- Bogleheads' Guide To Investing
- Capitulation
- Risk and return: an introduction
- Risk and return: application

References

1. Reilly (1994), p.11
2. Bodie, Kane, Marcus, 2008, p. 5
3. "Annual Returns on Stock, T.Bonds and T.Bills: 1928 - Current" (https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html). Damodaran Online.
4. Rick Ferri (2006). *All About Asset Allocation*. McGraw-Hill. p. 23. ISBN 978-0071429580.
5. Elton et al, 2003, p. 49
6. Zvi Bodie; Paula H. Hogan. "For Long-Term Investors, the Focus Should Be on Risk" (https://www.aaii.com/features/jrnl200506p16.pdf) (PDF). AAII Journal. Retrieved October 17, 2023.
7. Gary Smith; Donald Gould. "Measuring and Controlling Shortfall Risk in Retirement" (https://web.archive.org/web/20190118052301/https://economics-files.pomona.edu/GarySmith/shortfallRisk.pdf) (PDF). Archived from the original (https://economics-files.pomona.edu/GarySmith/shortfallRisk.pdf) (PDF) on January 18, 2019.
8. Bodie, Kane, Marcus, 2008, p. 180
9. John Y. Campbell; Robert J. Shiller; Luis M. Viceira (2009). "Understanding Inflation-Indexed Bond Markets" (https://www.nber.org/programs/ag/rrc/09-20%20Campbell,%20Shiller,%20Viceira%20FINAL.pdf) (PDF). Retrieved April 1, 2012.

   > We view the experience with TIPS yields after the Lehman bankruptcy as the sign of a highly abnormal market situation, where liquidity problems suddenly created severe financial anomalies.

   [As a further reference, this can be seen directly on a price chart like the following: https://www.google.com/finance?chdnp=0&chdd=1&chds=1&chdv=1&chvs=Logarithmic&chdeh=0&chfdeh=
10. Bodie, Kane, Marcus, 1996, p. 173
11. Bodie, Kane, Marcus, 2008, p. 123
12. Reilly, p. 14
13. Bodie, Kane, Marcus, 2008, p. 124
14. Ilmanen, 2011, p.7
15. Bodie, Kane, Marcus, 2008, p.121
16. Reilly, p.14
17. Elton et al (2003), p.45
18. Markowitz, Harry (March 1952). "Portfolio Selection" (https://www.math.ust.hk/~maykwok/courses/ma362/07F/markowitz_JF.pdf). *The Journal of Finance* 7 (1): 77-91. Retrieved April 12, 2012.
19. Bodie, Kane, Marcus, 2008, p. 122
20. Elton et al (2003), p. 48

21. "Standard Deviation and Sharpe Ratio Methodology" (https://corporate.morningstar.com/us/documents/MethodologyDocuments/MethodologyPapers/StandardDeviationSharpeRatio_Definition.pdf) (PDF). Morningstar. 2005. Retrieved August 10, 2012.
22. Ilmanen, 2011
23. Bodie, Kane, Marcus, 2008, p. 112
24. Reilly, pp. 11-13
25. Swedroe (2002), *Rational Investing in Irrational Times*, pp. 191-200. Truman Talley Books, St. Martin's Press, New York
26. Ilmanen, p.32
27. Rick Ferri. "The Portfolio Solutions 30-Year Market Forecast For 2012" (https://web.archive.org/web/20121027031840/https://www.portfoliosolutions.com/the-portfolio-solutions-30-year-market-forecast-for-2012/). Archived from the original (https://www.portfoliosolutions.com/f-13.html) on October 27, 2012.
28. Ed Tower (November 10, 2008). "Are GMO's Predictions Prescient? Using them to predict Vanguard's Mutual Fund Returns" (https://public.econ.duke.edu/Papers//PDF/GMO_Predictions1.pdf) (PDF). Retrieved October 17, 2023.
29. Bodie, Kane, Marcus, 2008, p. 137
30. Reilly, 1994, p. 245
31. Bodie, Kane, Marcus, 1996, p. 197
32. "International Equity: Considerations and Recommendations" (https://institutional.vanguard.com/iwe/pdf/FASIECR_112006.pdf) (PDF). Vanguard Investment Counseling & Research. 2006. Retrieved August 10, 2012.
33. Bodie, Kane, Marcus, 1996, p 244
34. Bodie, Kane, Marcus, 1996, p. 267
35. Elton et. al., 2003, pp. 132-133
36. "Morningstar MPT Statistics Methodology" (https://corporate.morningstar.com/us/documents/MethodologyDocuments/MethodologyPapers/MorningstarMPTStatistics_Methodology.pdf) (PDF). Morningstar. 2005. Retrieved August 10, 2012.
37. Bodie, Merton, 2000, p. 259

Reading

- Frank K. Reilly, (1994). *Investment Analysis and Portoflio Management*. The Dryden Press, Harcourt Brace College Publishers, Fourth Edition. ISBN 978-0030970528.
- Zvi Bodie, Alex Kane, Alan J. Marcus, (2008). *Essentials of Investments*. McGraw-Hill. ISBN 978-0071263245.
- Edwin J. Elton, Martin J. Gruber, Stephen J. Brown, William N. Goetzmann, (2003). *Modern Portfolio Theory and Investment Analysis*. Wiley. ISBN 978-0471238546.
- Antii Ilmanen, (2011). *Expected Returns*. Wiley. ISBN 978-1-119-99072-7.
- Zvi Bodie, Robert C. Merton, (2000). *Finance*. Prentice-Hall. ISBN 978-0133108972.
- William Bernstein (2010), *The Investor's Manifesto*. John Wiley & Sons, Inc. ISBN 978-0470505144.
- Zvi Bodie, Alex Kane, Alan J. Marcus, (1996). *Investments*. McGraw-Hill. ISBN 978-0256146387.

External links

- Part of an online finance course that provides more in-depth coverage of much of the material in this Wiki article: Risk and Return (https://web.archive.org/web/20200227140918/https://ci.columbia.edu/ci/premba_test/c0332/s6/index.html). *PreMBA Finance*, Columbia University. (Archive copy.)
- The file used to generate the Figures is available as a free download from Google Docs: Risk - Historical Performance of Bonds and Stocks.xlsx (https://docs.google.com/open?id=0B8VSNcUR-B6Idk9welFiMDhRX3UybzZxdjdseDVwdw).

Bibliography

- Zvi Bodie (December 1994). "On The Risk of Stocks in the Long Run (https://ssrn.com/abstract=5771)". Harvard Business School Working Paper No 95-013. Retrieved 30 March 2012.
- Zvi Bodie, Jonathan Treussard, and Paul Willen, (May 2007). "The Theory of Life-Cycle Saving and Investing (https://ssrn.com/abstract=1002388)". FRB of Boston Public Policy Discussion Paper No. 07-3. Retrieved 30 March 2012.
- Elroy Dimson, Paul Marsh, and Mike Staunton, (February 2002). "Long-Run Global Capital Market Returns and Risk Premia (https://ssrn.com/abstract=299335)". London Business School Accounting Subject Area No. 035. Retrieved 30 March 2012.
- Elroy Dimson, Paul Marsh, and Mike Staunton, (December 2003). "Irrational Optimism (https://ssrn.com/abstract=476981)". LBS Institute of Finance and Accounting (Working Paper No. IFA397). Retrieved 3 April 2012.

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